Overview: energy-conditioned vacancy generation

Adaptive labels, latent information, and entropy sampling

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Observed-range labels improve first-round balanced sampling by roughly six percentage points, but a second acquired batch does not sustain that gain against its matched continuation. The follow-up recovers a similar gain by relabeling H0 alone. This identifies conditioning and its induced class weighting as important parts of the result; adding configurations is not a uniform benefit. Most of the gain is in HIGH-class fidelity, rather than low-energy discovery.

The tested CVAEs do not establish a primary sampling advantage over the no-latent autoregressive model. Their deterministic posterior means retain a recoverable vacancy-ordering fingerprint. A fixed-condition random-encoder control also contains this fingerprint, so recovery alone does not show that training discovered it. That information does not become reliable control of generated structures. In the illustrated fixed condition, tiny mean variations coexist with nearly unit-scale posterior noise; a readable mean scatter and a useful sampled latent are different outcomes.

Entropy estimation needs a different sampling objective. On the complete 12-site lattice, correct orbit multiplicities and proposal probabilities are essential. Training toward a 1,000 K Boltzmann target reduces the equal-composition entropy MAE from 0.01310 to 0.00420 k_B/cell for UMA and from 0.01371 to 0.00313 for MACE. The same proposals are worse than uniform sampling at 5,000 K. Histogram flattening also does not establish energy-bin convergence in the Wang–Landau calculation.

These findings concern fixed O3-derived vacancy lattices and the supplied MLIP energies. Agreement across teachers is useful sensitivity evidence; it does not validate DFT accuracy, relaxed phase stability, bulk entropy, or molecular-dynamics predictions.

1 What is generated and how success is measured

We generate lithium-vacancy patterns in a fixed layered $LiCoO_2$ parent. Patterns with the same lithium count can have different energies. Sampling selected energy ranges could broaden datasets for later interatomic-potential fitting. A configuration is a 48-entry binary vector $\mathbf{x}$: one means a lithium atom occupies that parent site; zero means a vacancy. The lithium count is fixed at 12, 24, or 36, corresponding to 25, 50, or 75% occupancy. The cobalt–oxygen framework, cell, and retained atomic positions stay fixed. The O3 label describes this layered parent and its three-layer stacking repeat. Relaxed structures and transitions to other stackings or phases are outside this dataset.

UMA-S-1.2.1/`omat` [1] supplies new training energies; the independent MACE-MH-1/`omat_pbe` teacher [2] evaluates the same candidates. These machine-learning interatomic potentials predict energy from an atomic structure. Their weights remain fixed while the generators are trained. No new density functional theory (DFT) calculations are run. Physical energy accuracy and thermodynamic stability consequently remain questions for later validation.

A periodic translation or retained point symmetry can turn one binary mask into another representation of the same pattern. All such masks form a symmetry orbit. We use the 192 retained N48 parent operations to assign each orbit a canonical identity. The starting H0 archive contains 40,157 TRAIN, 4,967 VAL, and 5,123 TEST records. Existing roles and protected identities are preserved;

newly generated development assessments are kept out of later acquisition. N48 training uses canonical masks; the entropy experiment later uses symmetry augmentation and multiplicity weights for labelled microstates.

The main statistic combines novelty and energy conditioning:

$$Y_{n,b} = \frac{\text{distinct eligible new orbits whose evaluated energy is in bin } b}{\text{all attempts at count } n \text{ and requested bin } b}. \tag{1}$$

Balanced yield averages equally over the nine count–bin cells. Duplicates, known/protected identities, and wrong-bin proposals stay in the denominator. The primary evaluation bins use the original teacher-specific H0 boundaries. Moving training bins are evaluated separately. Different assessment phases have different novelty histories, so their absolute yields are compared only within their common frozen history.

2 Adaptive energy labels: does iteration help?

An autoregressive (AR) decoder chooses sites sequentially, using the already chosen occupancies, conditions, and neighboring-site summaries. It enforces the lithium count exactly and draws each free occupancy from its predicted probability. A CVAE adds a Gaussian encoder and a latent vector z to that decoder [4]. Both receive composition and a requested LOW/MIDDLE/HIGH energy class. The graph encoder uses fixed parent chemistry and distances; the decoder separates three Li-site distance shells. These models generate occupancies, rather than energies or relaxed coordinates; each counted success requires an MLIP evaluation.

Following the iterative generate–evaluate–retrain idea of Wakamiya and Ishikawa [3], we compare three TRAIN-only label rules: original fixed cutoffs; updated 20th/80th percentiles; and observed-range cutoffs at $E_{\min} + 0.2R$ and $E_{\min} + 0.8R$, with $R = E_{\max} - E_{\min}$ within each composition. Quantiles describe the acquired distribution. Range labels emphasize observed extrema and can create very small classes. Neither rule knows the global extrema or makes generation follow an energy gradient.

Three paired seeds and both AR/$D = 2$ CVAE are followed through two rounds. Acquisition allows 512 attempts and at most 256 first eligible unique queries per cell. Every successful selected UMA evaluation enters TRAIN with its measured label. All three policies share the exact first acquired batch. Each warm start receives 2,000 updates and an equal-update continuation with no new batch. In round two the control begins at the same first-round lineage. Fresh assessments allow 128 attempts per cell. Uniform fixed-count mask sampling and fixed-seed mutation provide additional controls.

Table 1: Change in balanced H0 fixed-bin yield versus an equal-update no-new-data continuation, in percentage points, mean ± sample SD over three paired fit/proposal seeds. Round 2 uses the same first-round lineage as its control. The rounds have separate assessment histories.

Model / rule	UMA r1	UMA r2	MACE r1	MACE r2
AR / fixed	-1.33 ± 1.61	$+1.39 \pm 0.53$	-1.50 ± 0.85	$+1.27 \pm 0.35$
AR / adaptive	-0.35 ± 1.51	$+0.61 \pm 0.60$	$+0.03 \pm 2.23$	$+0.14 \pm 0.90$
AR / range	$+6.57 \pm 2.36$	-5.41 ± 1.48	$+6.13 \pm 1.35$	-6.31 ± 1.65
CVAE $D = 2$ / fixed	-0.90 ± 0.88	-0.17 ± 0.68	-0.69 ± 1.00	-0.55 ± 0.56
CVAE $D = 2$ / adaptive	$+0.17 \pm 1.61$	$+0.43 \pm 0.63$	$+0.12 \pm 1.66$	$+0.26 \pm 0.77$
CVAE $D = 2$ / range	$+6.08 \pm 1.97$	-6.28 ± 1.76	$+6.16 \pm 1.25$	-6.22 ± 1.48

Range labels improve the first-round balanced statistic by about six points under both teachers, but the second acquired batch reverses that gain relative to the appropriate continuation. The first-round same-data range-minus-fixed contrast is $+7.90 \pm 0.83$ points for AR and $+6.97 \pm 2.74$ for $D = 2$ under UMA. Most of that gain is HIGH-class conditioning: AR's fixed-LOW change is only $+0.35$ points,

whereas its HIGH change is +29.77 points. A broad-class gain is not an equally large gain in low-energy discovery.

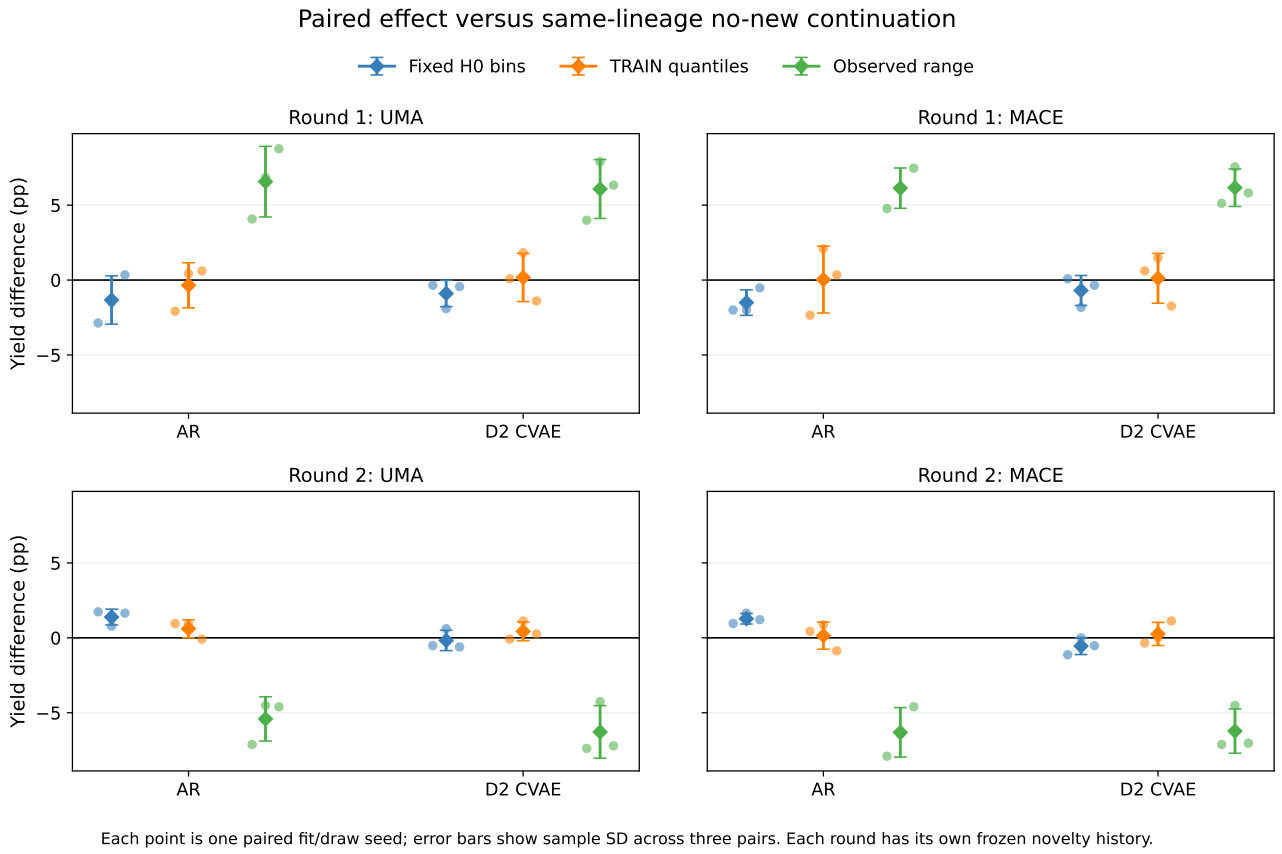


Figure 1: Acquisition effects within separately frozen assessments. Each round uses its appropriate no-new-data continuation; round 2 starts both arms from the same first-round lineage. Error bars are SD over three paired fit/proposal seeds.

After round two, absolute UMA range yields remain 70.28% for AR and 68.66% for $D = 2$, above the declared fresh uniform and fixed-TRAIN-seed mutation baselines of $33.88 \pm 0.59\%$ and $58.16 \pm 1.53\%$. Nevertheless, continuing the first-round model without the new range batch performs better. The decline mainly appears in fixed-bin condition fidelity; novelty still exceeds 97%.

This requires care in interpretation. Acquiring higher-energy extrema raises the range-defined LOW cutoff, so the requested target itself becomes less strict relative to the fixed-H0 LOW reference. The negative fixed-metric change and the positive extension of upper-energy coverage describe different outcomes. The detailed report separates novelty, condition fidelity, moving bins, endpoint growth, and loss-weight concentration rather than reducing them to one success label.

A fresh, explicitly post-hoc control shows that most of the first-round range-label gain does not require new structures. Starting from the same original checkpoint and allocating the same 2,000-update continuation, relabeling H0 alone improves balanced UMA yield by 6.48 ± 1.01 points for AR and 7.06 ± 1.09 for $D = 2$. Independent MACE assessment gives 6.11 ± 0.98 and 6.80 ± 0.91 points. At identical first-round range edges, adding the acquired rows instead changes UMA yield by -1.36 ± 0.39 points for AR and $+1.85 \pm 1.33$ for $D = 2$; MACE gives -2.05 ± 0.84 and $+2.00 \pm 1.39$ points. These are three-pair sample SDs.

The larger improvement is therefore available through conditioning and its associated loss reweighting on existing data. The smaller added-row effect depends on architecture. This does not negate the

Table 2: Posthoc conditioning/data controls: balanced H0 fixed-bin yield change, percentage points, mean ± sample SD over three paired fits. Relabeling also changes class-balanced loss weights. The matched-range comparison holds the physical edges identical; adding rows can still change minibatches and row weights.

Model / contrast	UMA	MACE
AR / H0 range versus fixed	$+6.48 \pm 1.01$	$+6.11 \pm 0.98$
AR / New rows / fixed edges	-1.56 ± 0.88	-1.10 ± 1.33
AR / New rows / matched range edges	-1.36 ± 0.39	-2.05 ± 0.84
CVAE $D = 2$ / H0 range versus fixed	$+7.06 \pm 1.09$	$+6.80 \pm 0.91$
CVAE $D = 2$ / New rows / fixed edges	$+0.41 \pm 0.78$	$+0.09 \pm 0.46$
CVAE $D = 2$ / New rows / matched range edges	$+1.85 \pm 1.33$	$+2.00 \pm 1.39$

measured expansion of upper-energy samples: fixed-bin fidelity and exploration of a wider observed energy range answer different questions. Neither outcome establishes DFT accuracy, physical stability, or global coverage.

3 Does the latent vector add useful physical control?

We test latent dimensions $D = 1, 2, 4$, a cyclic KL schedule [5], and genuine, condition-only, and shuffled physical-target heads. During training the encoder supplies a mean μ and standard deviation σ, and samples $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, multiplying coordinates element by element. Generation uses a standard-normal prior without an input mask. The Kullback–Leibler (KL) penalty encourages agreement between these latent distributions. The target C_{vv} is the fraction of nearest-neighbor periodic edges whose two sites are vacant. It describes local vacancy clustering. The auxiliary target is centered and scaled using TRAIN data within each composition–energy-bin cell, so conditions alone cannot explain its mean variation. Genuine-head weights 1, 10, and 100, plus supervision of the posterior mean instead of its noisy sample, are tested with matched controls. The latent studies complete 85 fits, all at 4,000 updates and five seeds per family. These matched budgets do not certify full optimizer convergence; the detailed report retains loss curves.

Table 3: Primary balanced novel correct-bin yield, percent, mean ± sample SD over five fit/proposal seeds. Compare arms within this phase's fresh development history.

Model	UMA	MACE
AR	62.92 ± 1.41	63.82 ± 0.99
CVAE $D = 1$	63.12 ± 1.40	63.68 ± 0.75
CVAE $D = 2$	62.76 ± 0.60	63.49 ± 1.38
CVAE $D = 4$	63.39 ± 1.75	64.03 ± 0.41

None of these latent interventions establishes a primary sampling advantage over AR. On a separate fresh assessment, mean supervision gives $63.33 \pm 1.76\%$ UMA yield versus $63.47 \pm 0.86\%$ for AR, with five-fit sample SD. Genuine guidance does not reliably move the actual generated vacancy-pair fraction in its requested direction; shuffled heads can move it as much or more. A small positive contrast under stronger supervision is retained in the detailed report, together with its unsuccessful AR and shuffled-target comparisons.

The representation contains more information than the original pooled probe suggested. A posthoc probe, standardized and fitted separately within each condition using TRAIN only, recovers clustering from VAL posterior means. Balanced MSE is 0.1009 ± 0.0963 for $D = 2$, versus 1.0212 for the zero reference. Yet random graph initialization recovers this property even more accurately in the fixed-condition control: physical graph structure already provides useful features before training.

This does not give useful sampled-latent control. In one predeclared 24-Li/MIDDLE illustration, variations of the mean are roughly 10^{-6}–10^{-8}, whereas posterior sampling noise has standard deviation about one. An analytic Gaussian-mixture calculation bounds information distinguishing the 986 empirical VAL records in noisy Z below 10^{-10} nats for the three illustrated fits. This concerns the encoder joint in one condition; prior-decoder interventions are a separate test. The detailed report supplies the derivation, high-precision checks, and corrected mean/noise plots. A colored mean scatter alone would therefore overstate useful physical information in generated samples.

4 Entropy: weighting the proposal for a different task

Novel-structure search and entropy estimation use different counting rules. For a new idealized 12-site O3 lattice, exhaustive enumeration gives 4,096 labelled occupancies and 87 symmetry orbits. At half occupancy, 924 labelled masks form only 17 orbits. If orbit o represents g_o masks, its thermal weight must include g_o:

$$Z_{\mathrm{rel}} = \sum_o g_o e^{-\epsilon_o/(k_{\mathrm{B}}T)}, \qquad S/k_{\mathrm{B}} = \log Z_{\mathrm{rel}} + \langle\epsilon\rangle/(k_{\mathrm{B}}T). \tag{2}$$

$\epsilon_o = E_o - E_{\min}$ is total energy per cell in eV, T is K, and k_{B} is Boltzmann's constant. This is finite-cell configurational entropy; vibrations, spin states, phase changes, and cell-size transfer are excluded. Each teacher evaluates 87 representatives plus 36 symmetry checks. The exact reference expands representative energies by multiplicity. Measured numerical symmetry errors, particularly UMA at low temperature, qualify this assumption.

A learned proposal $q(x)$ is not automatically a thermal distribution. Importance sampling divides each Boltzmann factor by the actual proposal probability [7]. Weight effective sample size (ESS), $(\sum_m w_m)^2/\sum_m w_m^2$, measures concentration of importance weights w_m; M is the number of draws. We use exact AR probabilities and a 10% uniform mixture to guarantee support, retain repeated masks, and compare against the complete small-lattice reference. The original composition-only AR is less efficient than uniform in the main comparison. A follow-up instead trains three same-seed AR models per teacher using TRAIN weights proportional to $g_o \exp[-E_o/(k_{\mathrm{B}} \times 1000\,\mathrm{K})]$, normalized to preserve each composition's original TRAIN weight mass. No held-out energy affects these weights; TEST energies remain NaN in learning exports.

Table 4: Temperature transfer of the 1,000 K-trained proposal. Entries are descriptive equal-composition averages over $n = 3, 6, 9$ at 8,192 draws; MAE is in k_{B} per cell and ESS/M is dimensionless. They are not averages of a physical entropy across compositions.

Teacher	T (K)	AR MAE	Weighted MAE	Uniform MAE	W. ESS/M	U. ESS/M
UMA	300	0.026879	0.008345	0.020461	0.3038	0.1044
UMA	1000	0.013095	0.004204	0.009409	0.7296	0.2495
UMA	5000	0.001973	0.008512	0.001763	0.6274	0.8259
MACE	300	0.020480	0.005252	0.016783	0.6374	0.2549
MACE	1000	0.013710	0.003133	0.011170	0.7420	0.3006
MACE	5000	0.001984	0.009692	0.001704	0.6515	0.8267

At 1,000 K, target weighting improves measured mean absolute entropy error (MAE) and weight ESS for all three tested compositions, but the same proposal performs worse than uniform at 5,000 K. The table reports descriptive averages across the three compositions, rather than a physical average entropy. Three fits and three draw seeds per fit provide the sampling comparisons. These draws reuse the tabulated energies, so they benchmark estimator error rather than additional MLIP cost. A CVAE would require its latent-marginal proposal probability; the decoder likelihood at one latent sample is insufficient for this weight.

There is also an important negative numerical result. All 36 bounded Wang–Landau runs [6] meet their specified schedule, but UMA's Li6/12 entropy at 300 K changes materially with energy-bin resolution: the exact discrete value is $3.26849\,k_{\mathrm{B}}$, versus exact midpoint-binned values 4.34857 and 3.51230 for 16 and 32 bins. Completing a flat-histogram schedule has not established bin-width convergence.

5 Decisions, reproducibility, and collaborator tasks

Retain AR as the reference generator and treat range relabeling as a conditioning/reweighting policy. Its gain does not establish that another acquired batch is beneficial, or that LOW requests improve as much as HIGH requests. Future comparisons should preserve a common assessment history and separate fixed-bin fidelity from extension of the observed range.

The next latent test should target information that survives posterior sampling and produces a directed change in generated structures. Recovering ordering from tiny encoder-mean variations is useful diagnosis, but does not establish that the learned latent improves generation. The entropy study likewise shows why the sampling objective matters: its 1,000 K proposal helps that target but loses to uniform sampling at 5,000 K.

- **Ricardo** reviews endpoint structures, symmetry and units, interprets teacher disagreements, and defines later DFT/relaxation checks.
- **Beihao** reproduces AR/CVAE training and develops a controlled sampled-latent test with the existing data, matched budgets, and genuine, condition-only, shuffled and random-encoder comparisons.
- **Varad** reproduces the paired evaluations, audits thresholds and denominators, reports per-cell variability and negative results, and maintains the analysis and cost receipts.

Independent DFT accuracy, relaxed structures, non-O3 phases, size transfer, and stable dynamics remain physical validation tasks. The current fixed-lattice MLIP results do not establish those outcomes.

The integrated package checks pass 414 tests, with six skips. Full task arguments, checkpoint/input hashes, raw proposals, exclusions, failed technical attempts, teacher receipts, and assessment boundaries remain in the run archive. The accompanying detailed report and compact evidence packet provide the numerical tables, derivations, figures, commands, and cost definitions. Neural/MLIP computation uses FP32 with TF32 disabled; entropy sums and moment diagnostics use float64. No new DFT, MD, or MLIP fine-tuning is performed. The campaign completes 166 generative fits (520,000 optimizer updates) and 469,672 geometry–teacher static evaluations, including repeat evaluations required by separately frozen comparisons. This is an evaluation-event count, not the number of globally unique physical geometries. The final allocation receipt reports all four Slurm allocations, including startup, idle time and failed technical subprocesses, with GPU-node-hours defined as allocated GPU-device hours divided by four. Shared-node allocations can overlap on the same physical node. This estimate is distinct from an official posted charge; the compact packet's `resource_receipt.json` records final usage of 28.8799 GPU-node-hours (115.5194 allocated GPU-device hours), below the 150 GPU-node-hour limit. All four allocations have ended; three idle allocations were explicitly released and the final shared workers exited normally.

References

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